

Carlos Mora Sardiña

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EDUCATION

Ph.D. in Computational Science and Engineering

University of California, Irvine

Irvine, California

Sep. 2021 – Dec. 2025

- Thesis: *Hybrid Gaussian Process and Deep Learning Frameworks for Computational Science and Engineering*

B.S. + M.S. in Aerospace Engineering

Polytechnic University of Catalonia

Barcelona, Spain

Sep. 2016 – June 2022

SKILLS

Programming & ML Libraries: Python, Matlab, C++, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Modeling & Applied ML: Deep Learning, Scientific ML, Gaussian Processes, Data Fusion, Uncertainty Quantification

MLOps & Infrastructure: Git, HPC, MLflow, Spark, Databricks, Airflow, A/B testing

EXPERIENCE

Graduate Student Researcher / Teaching Assistant

University of California, Irvine

Irvine, California

Sep. 2021 – Dec. 2025

- Developed an operator learning framework based on Gaussian processes that outperforms other neural operators while using fewer parameters. It is the first zero-shot learning mechanism for operator learning in the literature.
- Built NN-CoRes, a physics-informed machine learning approach that integrates neural networks with kernel methods for solving PDEs. NN-CoRes achieves enhanced accuracy and robustness over state-of-the-art methods.
- Designed Pro-NDF, a probabilistic machine learning method for data fusion based on Bayesian neural networks. Pro-NDF enables the integration of an arbitrary number of data sources to enhance the prediction accuracy and reliability at a lower data acquisition cost.
- Contributed to GP+, an open-source Python library built on PyTorch for machine learning and statistical modeling via Gaussian processes. GP+ systematically integrates multi-fidelity emulation and Bayesian optimization.

Machine Learning Engineering Intern

Patreon

San Francisco, California

June 2025 – Sep. 2025

- Built and optimized an end-to-end machine learning pipeline for improving creator discovery surfaces, resulting in improved recommendation relevance and user engagement.
- Performed exploratory data analysis and trained, validated, and iterated on machine learning models to evaluate new recommendation ideas.
- Deployed machine learning models to production and developed observability tools to monitor model performance.

Software Engineer Intern

Applus+ Laboratories

Barcelona, Spain

March 2019 – July 2019

- Developed the software components in C++ and MATLAB for a test bench aimed at evaluating the performance of the main gearbox in helicopters.
- Automated processes using programmable logic controllers, ensuring a smooth and efficient test bench operation.

AWARDS AND HONORS

Full scholarship, Bellsells fellowship

Given to students with high academic performance to pursue graduate studies in the United States.

Sep. 2021

Honorable mention, Catalan Government

Top 0.3% students in the university entrance exam in Catalonia.

June 2016

SELECTED PUBLICATIONS

- **Mora, C.**, Yousefpour, A., Hosseinmardi, S., Bostanabad, R. (2024). “Integrating Kernel Methods and Deep Neural Networks for Solving PDEs” *ICLR 2024 (Workshop on AI4DifferentialEquations In Science)*.
- **Mora, C.**, Eweis-Labolle, J. T., Johnson, T., Gadde, L., Bostanabad, R. (2023). “Probabilistic Neural Data Fusion for Learning from an Arbitrary Number of Multi-fidelity Data Sets” *Computer Methods in Applied Mechanics and Engineering*.